

DONALD AADITHIYAN

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Wellewatte, Colombo - 06, Sri Lanka

SUMMARY

Final-year undergraduate in Computer Science and Engineering (Data Science stream) at the University of Moratuwa. Research interests span reinforcement learning, model-based RL, multi-agent systems, uncertainty quantification, and interpretability, with a focus on understanding how AI systems make decisions, fail under uncertainty, and generalize in complex environments. Interested in trustworthy decision-making for real-world deployment, counterfactual reasoning, and AI for societal-scale systems. Currently studying epistemic uncertainty in DreamerV3 world models and investigating how internal latent representations relate to planning reliability and failure detection. Also developing interpretable machine learning architectures for structured decision-making tasks. Proficient in Python, PyTorch, and the full ML pipeline.

EDUCATION

- **University of Moratuwa** Mar 2022 – Present
Bachelor of Science (Honours) in Engineering Moratuwa, Sri Lanka
 - Specialization: Computer Science and Engineering
 - Stream: Data Science and Engineering
 - Current CGPA: 3.54/4.0 (Second Upper Class)
 - Relevant Coursework: Deep Neural Networks, Calculus, Operational Research, Data Structures & Algorithms, Database Systems
- **St. Peter's College** Jan 2019 – Jan 2022
GCE Advanced Level Colombo, Sri Lanka
 - Subjects: Distinction(A) in Physics, Chemistry, Combined Mathematics
 - Z-score: 2.2108 (Top 1.67% in Sri Lanka)
- **St. Peter's College** Jan 2017 – Dec 2018
GCE Ordinary Level Colombo, Sri Lanka
 - Results: 9 A's

PUBLICATIONS

C=CONFERENCE, W=WORKSHOP (PEER-REVIEWED), I=IN PROGRESS

- [C.1] Donald Aadithiyan, Birunthaban Rajendram, and Parishith Ragumar. (2025). **Predicting YouTube Video Initial Engagement Using Ensemble Machine Learning Techniques**. In *Proceedings of the International Conference in Data Science 2025*. Department of Statistics, University of Colombo, Sri Lanka.
Link:  [Research Paper](#)
- [W.1] Donald Aadithiyan (2026). **Beyond Point Prediction: Inter-Temporal Conditioning and Conformal Uncertainty for Multi-Horizon Rare-Event Forecasting**. Submitted to *AI for Forecasting Workshop, International Conference on Machine Learning (ICML) 2026*. (Under Review — Decision Pending)
- [I.1] Donald Aadithiyan. (2026). **When Better Models Predict Less: KAN Decoder Curvature as a Regime Attribution Signal on the JSE**. *Diff Capital, Johannesburg*. (Working Paper — In Progress)

RESEARCH PROJECTS

- **Epistemic Uncertainty in RL World Models — Phase 1 & 2** Independent Research
Tools: Python, PyTorch, DreamerV3 (XS), Block GRU, linear probing, causal tracing
 - Probed whether DreamerV3's recurrent hidden state h_t linearly encodes epistemic uncertainty without external OOD labels; confusion signal shown to be orthogonal to input novelty.
 - Uncertainty representation uniformly distributed across all GRU blocks with no localised subspace — structurally distinct from ROME-style localisation in transformer FFN layers; causal tracing framework extended to recurrent GRU dynamics.
 - Phase 2 pilot: probe adds growing predictive information at longer temporal horizons beyond scalar KL autocorrelation, confirming trajectory-level confusion context in h_t .
- **CAE-KAN: Interpretable Equity Valuation with KAN Decoder** Research — Differential Capital, Johannesburg
Tools: Python, PyTorch, KAN (B-spline), FiLM conditioning, sparse autoencoders, JSE data
 - Designed CAE-KAN — a Conditional Autoencoder with KAN decoder mapping latent valuation factors through learnable B-splines; introduced Valuation Curvature Signal (VCS) as a per-factor OOD stress measure attributing 8/8 JSE market regime events to their economic transmission mechanisms.

- Formalised the accuracy-signal tradeoff: accurate valuation models structurally absorb the mean-reverting value premium on thin emerging markets, reframing interpretability as the correct benchmark.
- Established fair value identity principle for multi-task learning; empirical mispricing-return response function reveals declining efficiency trend decomposed into model conservatism and genuine market efficiency.
- **Beyond Point Prediction: Multi-Horizon Rare-Event Forecasting** Independent Research
Tools: Python, Prophet, ARIMA, XGBoost, LightGBM, CatBoost, ElasticNet, Conformal Inference, SHAP, FRED API
 - Built a two-stage decoupled pipeline: hybrid Prophet/ARIMA with XGBoost residual correction, followed by a stacking ensemble RegressorChain conditioning longer-horizon recession probabilities on shorter-horizon outputs.
 - Novel finding: calibration set rare-event class composition — not size — determines conformal interval validity under temporal distribution shift; first empirical demonstration in regression rare-event forecasting.
 - Decoupled architecture enables counterfactual stress testing without retraining; horizon-conditional SHAP reveals feature importance shifting from current-state to forward-looking signals across horizons.
- **YouTube Viewership Forecasting Tool** Personal Project
Tools: Python, YouTube Data API v3, XGBoost, LightGBM, CatBoost, Random Forest, Streamlit, GCP
 - Collected hourly metadata from 10,000+ YouTube videos on GCP; engineered engagement scores and log-transformed distributions.
 - Implemented a two-stage pipeline: classification model predicting engagement tiers followed by tier-specific quantile regression; ensemble stacking outperformed all individual baselines.
 - Deployed via an interactive Streamlit dashboard for content creators.

PROFESSIONAL EXPERIENCE

- **Differential Capital PTY LTD** Dec 2025 – Present
Data Science Intern (Industrial Training) & Research Collaborator  [Differential Capital](#)
 - Developed and maintained ETL pipelines for retail webscraper infrastructure; built monitoring dashboard tracking data quality metrics and pipeline health.
 - Conducted EDA on historical scraped data, documenting schema inconsistencies and improving webscraper robustness with enhanced error handling and validation.
 - **Research:** Developing CAE-KAN for interpretable equity valuation on the JSE; introduced Valuation Curvature Signal (VCS) for per-factor regime attribution. (Working paper in preparation)
- **Rising Talent Freelance Full Stack Developer – Upwork** Jan 2025 – Feb 2026
Rising Talent Badge - Top 15% in the Industry
 - Delivered two full-stack apps (Flutter, Next.js, Firebase) to international clients with 5-star feedback.
 - **Moon App** – Flutter app connecting and controlling a smart design object via BLE. **Links:** [Android](#) | [iOS](#)
 - **Fixsy** – Service booking app and admin panel (Flutter, Next.js, Firebase) with real-time analytics dashboard for booking patterns and technician performance metrics.
- **Collaborative Development Initiative – Lightspeed Labs** Dec 2024 – Mar 2026
Project Manager and Full Stack Developer  [Lightspeed Labs](#)
 - Led a peer-driven team delivering full-stack web and mobile solutions for local and international clients. Earned a revenue of over 1.3 Million LKR.
 - **Social Pass** – Event discovery app with real-time attendance tracking and predictive analytics for event popularity. **Links:** [Android](#) | [iOS](#)
 - **Grossoo** – E-commerce platform (Flutter, MedusaJS) with customer behavior analytics and demand forecasting. **Link:** [Android](#) | [iOS](#)

HACKATHON PROJECTS

- **Localize Sri Lanka** Team Project
Tools: Flutter, Firebase, OpenAI API [Project link](#)
 - Developed a comprehensive tourism platform where guides can list travel packages and share content (images, videos) in a social media-like feed, while tourists can book packages and access assistance via an AI-powered chatbot.
 - Integrated OpenAI for conversational support, Firebase for real-time updates and authentication, and Flutter for cross-platform mobile deployment.
 - Enabled tourists to apply for visas, book accommodations, and explore curated content in one unified app.
 - A project that won two major national hackathons: **IDEALIZE 2024** and **DevThon 2024**.

SKILLS

- **Programming Languages:** Python, R, SQL, Dart, TypeScript, JavaScript, Java, C++
- **ML & Deep Learning:** PyTorch, Scikit-learn, Reinforcement Learning (PPO, DreamerV3), Probing & Interpretability, Conformal Inference, SHAP, Ensemble Methods, Time Series Analysis
- **Optimization & RL:** Deep RL, Model-Based RL, RegressorChain, Optuna, Multi-Horizon Forecasting, Combinatorial Optimization
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, Statistical Modeling, Causal Tracing
- **Web & Mobile Development:** Flutter, Next.js, React.js, Node.js, FastAPI, Laravel
- **Database & Cloud:** MySQL, PostgreSQL, Firebase, Supabase, GCP
- **Development Tools:** Git, REST APIs, Data Visualization, CodeMagic
- **Soft Skills:** Research Writing, Data Storytelling, Client Communication, Project Management, Problem Solving

HONORS AND AWARDS

- **Winners of IDEALIZE 2024** 2024
Project: Localize Sri Lanka [!\[\]\(0551a83d441798e532995956b603f604_img.jpg\) Post Link](#)
 - A prestigious nationwide mobile and web development competition with participation from over 200+ teams across the country.
 - **Double Victory:** Secured the **Idealizer Award** for best overall team performance and the **Trendsetter Award** recognising our innovation and national impact.
- **Winners of DevThon 2024** 2024
Project: Localize Sri Lanka [!\[\]\(54ee180c0037b66a36ce2219a481afde_img.jpg\) Post Link](#)
 - A prestigious nationwide mobile and web development competition organised by Leo Club of University of Moratuwa with participation from over 100+ teams across the country.
- **4th Place – Startup Spark 2024** 2024
Project: Localize Sri Lanka [!\[\]\(73ae654e8897db9b21f1bf9d9efc07ef_img.jpg\) Post Link](#)
 - Secured 4th place in a national competition organised by the IESL Student Chapter of University of Moratuwa during Techno 2024, with 50+ participating teams.
 - Pitched our startup to a panel of investors, highlighting business potential, product-market fit, and vision.
 - Gained hands-on experience in startup storytelling, confident pitching, and building investor trust.

EXTRA-CURRICULAR ACTIVITIES

- **Member – MoraSpirit Web & Technology Pillar** 2024 – 2026
University of Moratuwa
 - Contributed as a Laravel backend developer to build and maintain the MoraSpirit Sports Website's admin panel — Sri Lanka's premier university sports platform.
 - Ensured real-time score updates, match schedules, and athlete data were accurately served to thousands of daily users.
- **Co-Chairperson – SpiritX 2025** 2025
University of Moratuwa
 - Led the planning and execution of SpiritX 2025, MoraSpirit's first-ever inter-university hackathon with 160+ teams and 650+ participants from 20+ universities across Sri Lanka.
 - Oversaw team operations, sponsor engagement, and judging coordination while fostering a spirit of innovation and collaboration.

ONLINE LEARNING & CERTIFICATIONS

- **Building Transformer-Based NLP Applications – NVIDIA (Certificate of Competency) :** [Certificate link](#)
- **Feature Engineering by Kaggle :** [Certificate link](#)

ADDITIONAL INFORMATION

Interests: Self-taught investor with active investments in the Sri Lankan stock market and treasury bill market for the past 2 years. Deeply passionate about finance, macroeconomics, and market analysis. Regular contributor to quantitative finance discussions and market prediction modeling.

Languages: English (Professional working fluent proficiency), Sinhala (Professional working fluent proficiency), Tamil (Native proficiency)